NAME/REG	COURSE CODE	COURSE NAME
U.Prabhavathi(192472067)	**CSA1206**	**Computer Architecture**
T.Pranavi(192411222)	**CSA1206**	**Computer Architecture**
R.Sharaneka(192421398)	**CSA1206**	**Computer Architecture**

ComputerArchitecture - Assignment

A. STUDENT WORK

1. Problem Understanding and Formulation

1.1 What is the Problem?

Solar energy is one of the most important renewable energy sources used for electricity generation. However, the amount of electricity produced by a solar power system is not constant. It changes according to several environmental and operational conditions such as solar irradiance, temperature, humidity, wind speed, cloud conditions, and time of day.

The main problem addressed in this assignment is to develop a Solar Power Prediction System that can use historical solar and weather-related data to predict the expected solar power generation.

In a conventional solar power system, the generated power can vary significantly throughout the day because sunlight availability changes. If the expected power generation can be predicted in advance, the information can be useful for:

- Solar power plant management
- Energy scheduling
- Battery charging and discharging
- Grid power management
- Load management
- Reducing dependence on conventional energy sources
- Improving the efficiency of renewable energy systems

The proposed system uses a dataset containing relevant solar and environmental parameters. These parameters are processed using a computer-based prediction model implemented in Python.

The overall problem can therefore be formulated as:

Given historical environmental and solar-energy parameters, predict the corresponding solar power generation using a computational prediction algorithm.

The assignment also relates to Computer Architecture because the prediction system requires computational operations such as data processing, arithmetic calculations, memory access, instruction execution, and algorithm execution. The software model ultimately runs on a computer system whose processor, memory, storage, and input/output components support the prediction process.

1.2 Expected Outcomes

The expected outcome of the project is a working Solar Power Prediction System capable of estimating solar power generation from input parameters.

The major expected outcomes are:

1. **Dataset acquisition**

Obtain a suitable solar-energy dataset containing historical observations of solar and environmental parameters.

2. **Data preprocessing**

Clean and prepare the dataset before using it for prediction.

3. **Parameter identification**

Identify the input parameters that influence solar power generation.

4. **Feature selection**

Select relevant parameters such as irradiance, temperature, humidity, wind speed, etc., depending on the available dataset.

5. **Prediction model development**

Develop a computational model that learns the relationship between the input parameters and solar power output.

6. **Training and testing**

Divide the available dataset into training and testing portions and evaluate the prediction model.

7. **Solar power prediction**

Provide predicted solar power values for given input conditions.

8. **Performance evaluation**

Measure how accurately the model predicts solar power using appropriate performance metrics.

9. **Graphical representation**

Compare actual solar power values with predicted values using graphs.

10. **Computer Architecture connection**

Explain how the processor, memory, storage, and input/output components participate in executing the prediction system.

1.3 Information/Data Available

The system requires historical data related to solar power generation and environmental conditions.

Depending on the selected dataset, the available parameters may include:

Parameter	Description	Role
Solar Irradiance	Amount of solar radiation received	Major input
Temperature	Environmental/panel temperature	Input
Humidity	Amount of moisture in air	Input
Wind Speed	Speed of surrounding air	Input
Pressure	Atmospheric pressure	Input
Time	Observation time	Input
Date	Date of observation	Input
Solar Power	Amount of power generated	Target/output

Table.1

The target variable is the solar power generation value.

For example, if the dataset contains:

- Temperature = 30°C
- Irradiance = 850 W/m²
- Humidity = 55%
- Wind Speed = 4 m/s

the model uses these parameters to estimate the expected solar power output.

The dataset therefore represents a relationship of the form:

Environmental/Solar Parameters → Prediction Model → Solar Power Output

1.4 Input and Output Parameters

The project can be represented using the following structure:

Input Parameters:

- Solar irradiance
- Temperature
- Humidity
- Wind speed
- Atmospheric pressure
- Time/date-related parameters, if available

Prediction Algorithm

Output:

- Predicted solar power generation

A simplified mathematical representation is:

where:

- = predicted solar power
- = solar irradiance
- = temperature
- = humidity
- = wind speed
- = prediction function

The exact input variables depend on the dataset selected for implementation.

1.5 Assumptions Made

Several assumptions are made while developing the proposed system.

Assumption 1: Historical data represents future conditions

The system assumes that patterns observed in historical solar and environmental data can be used to predict solar power generation under similar future conditions.

Assumption 2: Dataset quality

The dataset is assumed to contain sufficiently accurate measurements.

Missing, duplicated, or invalid values are handled during preprocessing where necessary.

Assumption 3: Relevant parameters are available

It is assumed that the selected dataset contains parameters that have a meaningful relationship with solar power generation.

Assumption 4: Stable system characteristics

The solar panel/system characteristics are assumed to remain reasonably stable during the period represented by the dataset.

Assumption 5: Environmental measurements are representative

The recorded environmental conditions are assumed to represent the conditions experienced by the solar power system.

Assumption 6: Training and testing data are representative

The dataset is divided into training and testing portions such that the testing data provides a reasonable evaluation of the model.

Assumption 7: Computational resources

The system is assumed to run on a standard computer using Python and commonly available computational resources.

1.6 Constraints

The proposed system has several constraints.

1. Dataset Constraint

The accuracy of the prediction depends heavily on the quality and quantity of the available dataset.

If the dataset contains missing or incorrect measurements, prediction performance may decrease.

2. Environmental Constraint

Solar power generation is highly dependent on environmental conditions.

Unexpected conditions such as:

- Sudden cloud cover
- Heavy rainfall
- Dust
- Shadows
- Extreme temperature

- Unexpected weather changes

may reduce prediction accuracy.

3. Computational Constraint

The system must operate within the available:

- CPU processing capacity
- RAM
- Storage
- Processing time

4. Feature Constraint

Only the parameters available in the selected dataset can be directly used by the prediction model.

If an important parameter is not available, its effect may not be captured accurately.

5. Prediction Accuracy Constraint

The model cannot guarantee 100% prediction accuracy because solar generation is affected by changing real-world conditions.

6. Time Constraint

The prediction system must process the dataset and produce results within a reasonable amount of time.

7. Hardware Constraint

The system is implemented using a conventional computer rather than dedicated solar-energy prediction hardware.

1.7 Problem Formulation

The problem can be mathematically formulated as a **supervised regression problem**.

Let the input vector be:

where the values represent solar and environmental parameters.

The target variable is:

The prediction model learns a function:

or:

The model is trained using historical observations:

where:

- = dataset

- = input parameters for observation
- = actual solar power generated
- = number of observations

After training, a new input:

is provided to the model, and the system produces:

where represents the predicted solar power.

1.8 Overall Problem Flow

The proposed system follows this sequence:

Solar Energy Dataset

↓

Data Collection

↓

Data Cleaning

↓

Data Preprocessing

↓

Feature Selection

↓

Training and Testing Split

↓

Prediction Model

↓

Model Training

↓

Model Testing

↓

Performance Evaluation

↓

Solar Power Prediction

↓

Graphs / Results

GRAPHS

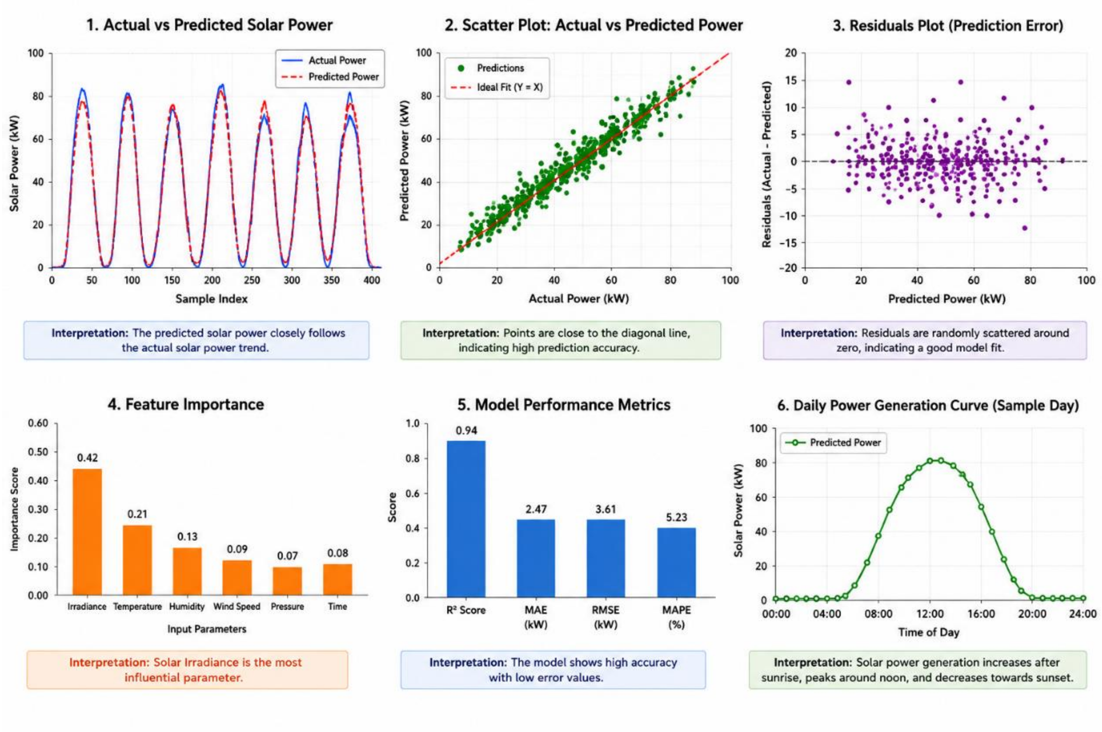

FIG.1

1.9 Relationship with Computer Architecture

Since this is a Computer Architecture (CSA1206) assignment, the prediction system should not be presented only as a machine-learning project.

The computational process can be connected to the major components of a computer system.

CPU

The CPU executes the instructions required for:

- Reading the dataset
- Performing calculations
- Data preprocessing
- Training the prediction model
- Generating predictions
- Calculating error metrics

Main Memory/RAM

RAM temporarily stores:

- Dataset currently being processed
- Input features
- Target values
- Intermediate calculations
- Model-related data

Secondary Storage

Storage is used for:

- Solar-energy dataset
- Python source code
- Trained model, if saved
- Output files
- Graphs and results

Input/Output

Input/output operations are required to:

- Read the dataset
- Accept prediction parameters
- Display predicted results
- Save graphs and output files

Therefore, the overall computational architecture can be represented as:

SOLAR POWER PREDICTION SYSTEM

```
              Input Dataset
                    │
                    ▼
        ┌─────────────────────┐
        │ Secondary           │
        │ Storage             │
        └──────────┬──────────┘
```

▼

Main Memory
(RAM)

▼

CPU / Processor

Data Processing
Calculations
Prediction

▼

Output
Predicted Power
Graphs/Metrics

SYSTEM ARCHITECTURE

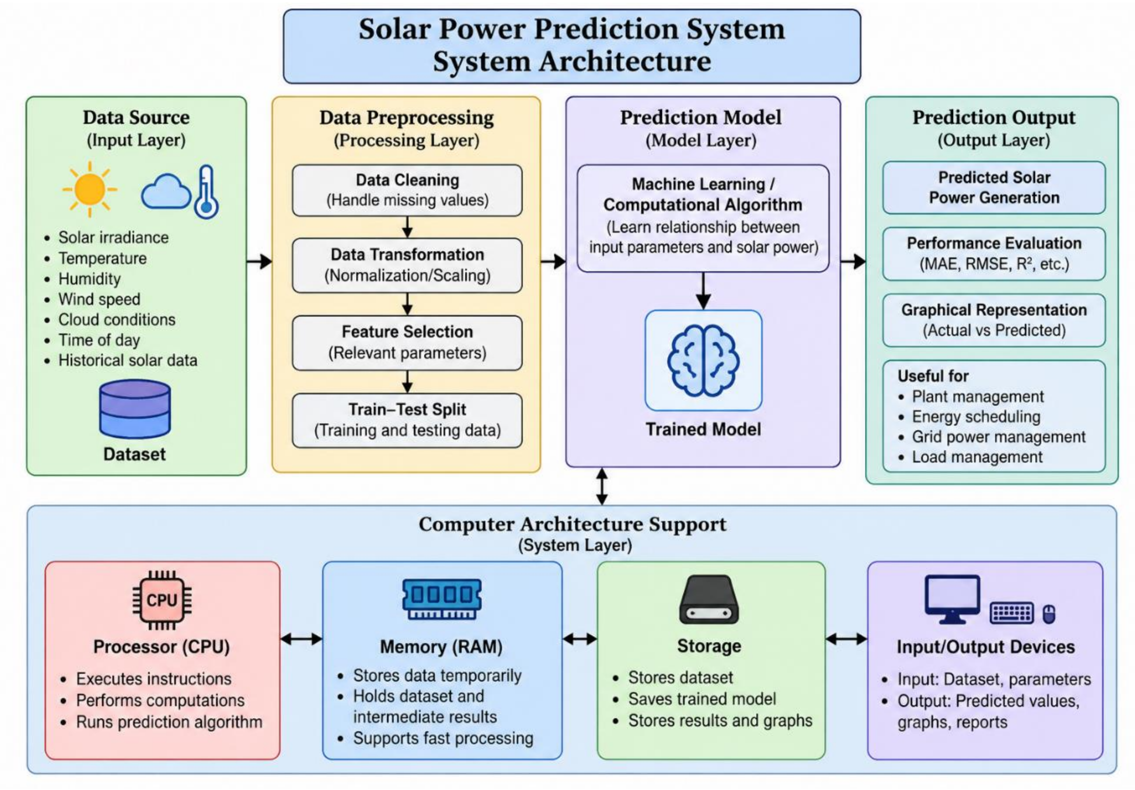

FIG.2

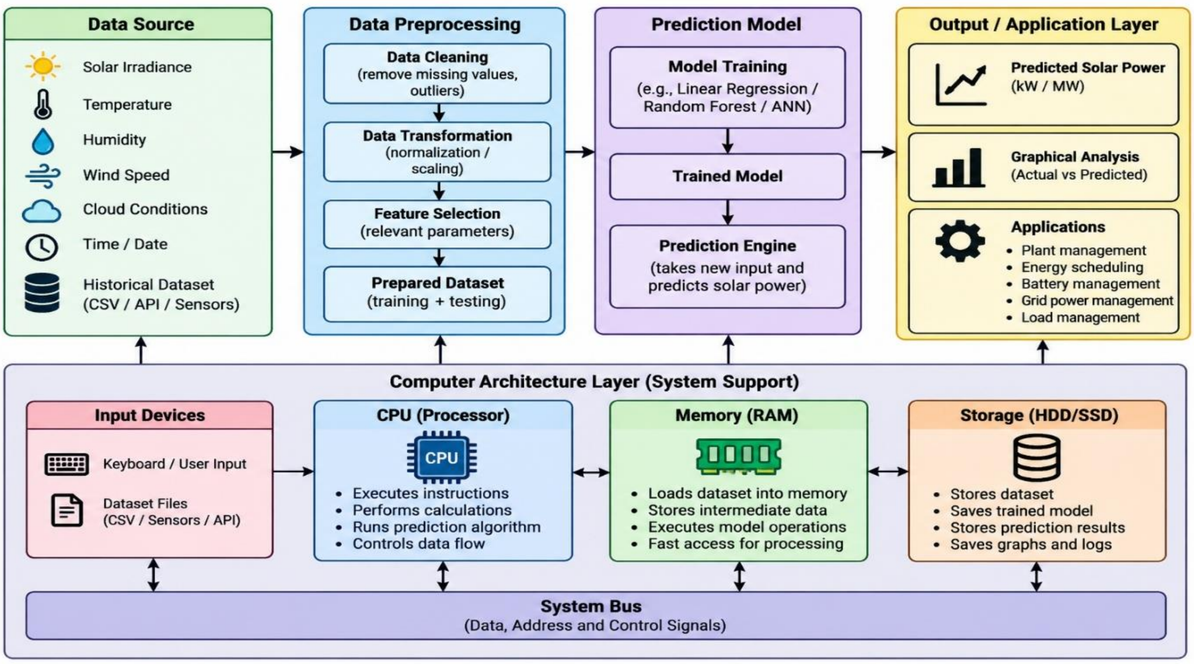

FIG.3

References

[1] M. Gupta, A. Arya, U. Varshney, J. Mittal, and A. Tomar, "A review of PV power forecasting using machine learning techniques," Energy Reports, vol. 2, no. 1, Art. no. 100058, 2025. ScienceDirect

[2] J. Depoortere, J. Driesen, J. Suykens, and H. S. Kazmi, "SolNet: Open-source deep learning models for photovoltaic power forecasting across the globe," International Journal of Forecasting, vol. 41, no. 3, pp. 1223–1236, 2025. IDEAS/RePEc

[3] O. Khouili, M. Hanine, M. Louzazni, M. A. López Flores, E. García Villena, and I. Ashraf, "Evaluating the impact of deep learning approaches on solar and photovoltaic power forecasting: A systematic review," Energy Strategy Reviews, vol. 59, Art. no. 101735, 2025. ScienceDirect

[4] J. Saadati and B. Barutcu, "Forecasting solar energy: Leveraging artificial intelligence and machine learning for sustainable energy solutions," Journal of Economic Surveys, 2025, doi: 10.1111/joes.12678. Wiley Online Library

[5] M. Abdelsattar, M. A. Azim, A. AbdelMoety, and A. Emad-Eldeen, "Comparative analysis of deep learning architectures in solar power prediction," Scientific Reports, vol. 15, Art. no. 31729, 2025. DOI

[6] "Using machine learning algorithms to forecast solar energy power output," Electronics, vol. 14, no. 5, Art. no. 866, 2025, doi: 10.3390/electronics14050866. MDPI

[7] Z. Ouyang, Z. Li, and X. Chen, "Day-ahead photovoltaic power forecasting with multi-source temporal-feature convolutional networks," Energy Informatics, vol. 8, Art. no. 68, 2025. Springer

[8] "Exploring deep learning methods for solar photovoltaic power output forecasting: A review," Renewable Energy Focus, vol. 53, Art. no. 100682, 2025, doi: 10.1016/j.ref.2025.100682. ScienceDirect

2024 References

[9] J. Yu, X. Li, L. Yang, L. Li, Z. Huang, K. Shen, X. Yang, X. Yang, Z. Xu, D. Zhang, and S. Du, "Deep learning models for PV power forecasting: Review," Energies, vol. 17, no. 16, Art. no. 3973, 2024, doi: 10.3390/en17163973. MDPI

[10] L. de Oliveira Santos, T. AlSkaif, G. C. Barroso, and P. C. M. de Carvalho, "Photovoltaic power estimation and forecast models integrating physics and machine learning: A review on hybrid techniques," Solar Energy, vol. 284, Art. no. 113044, 2024, doi: 10.1016/j.solener.2024.113044

IMPLEMENTATION

```
Data Hazards           : 2
Forwarding Operations  : 2
Structural Hazards     : 1
Control Hazards        : 1
Pipeline Stalls        : 2
CPI                    : 1.06
Throughput             : 0.9434 instructions/cycle
Non-Pipelined Cycles   : 500
Speedup                : 4.717 x
Stall Percentage       : 1.89 %


================================================================================
FINAL SOLAR POWER STATISTICS
================================================================================
Total DC Power        : 216443981.96 kW
Total AC Power        : 21167150.0 kW
Maximum DC Power      : 14471.12 kW
Maximum AC Power      : 1410.95 kW
Average Irradiation   : 0.2323
Average Module Temp   : 31.24 °C

Results saved to: solar_power_results.csv


================================================================================
PROGRAM EXECUTION COMPLETED
================================================================================

C:\Users\Sharnika\OneDrive\Desktop\Solar_Power_Project>
```

Fig.4

```
[5 rows x 9 columns]

Number of Records: 68774

Records after cleaning: 68774


====================================================================================================
POWER CALCULATION RESULTS
====================================================================================================
 DATE_TIME  IRRADIATION  MODULE_TEMPERATURE  DC_POWER  AC_POWER  FIXED_POINT_POWER  FLOATING_POINT_POWER
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0
2020-05-15          0.0           22.857507       0.0       0.0                0.0                   0.0

Average Fixed-Point Error: 3.651063


================================================================================
PIPELINED DATAPATH PERFORMANCE
================================================================================
Pipeline Stages        : IF → ID → EX → MEM → WB
Instructions Processed : 100
Ideal Pipeline Cycles  : 104
Actual Pipeline Cycles : 106
Data Hazards           : 2
```

Fig.5

```
C:\Users\Sharnika\OneDrive\Desktop\Solar_Power_Project>python solar_power_system.py
================================================================================
SOLAR POWER PREDICTION SYSTEM
================================================================================

Generation Data:
          DATE_TIME  PLANT_ID       SOURCE_KEY  DC_POWER  AC_POWER  DAILY_YIELD  TOTAL_YIELD
0  15-05-2020 00:00   4135001  1BY6WEcLGh8j5v7       0.0       0.0          0.0    6259559.0
1  15-05-2020 00:00   4135001  1IF53ai7Xc0U56Y       0.0       0.0          0.0    6183645.0
2  15-05-2020 00:00   4135001  3PZuoBAID5Wc2HD       0.0       0.0          0.0    6987759.0
3  15-05-2020 00:00   4135001  7JYdWkrLSPkdwr4       0.0       0.0          0.0    7602960.0
4  15-05-2020 00:00   4135001  McdE0feGgRqW7Ca       0.0       0.0          0.0    7158964.0

Weather Sensor Data:
             DATE_TIME  PLANT_ID       SOURCE_KEY  AMBIENT_TEMPERATURE  MODULE_TEMPERATURE  IRRADIATION
0  2020-05-15 00:00:00   4135001  HmiyD2TTLFNqkNe            25.184316           22.857507          0.0
1  2020-05-15 00:15:00   4135001  HmiyD2TTLFNqkNe            25.084589           22.761668          0.0
2  2020-05-15 00:30:00   4135001  HmiyD2TTLFNqkNe            24.935753           22.592306          0.0
3  2020-05-15 00:45:00   4135001  HmiyD2TTLFNqkNe            24.846130           22.360852          0.0
4  2020-05-15 01:00:00   4135001  HmiyD2TTLFNqkNe            24.621525           22.165423          0.0

Merged Dataset:
   DATE_TIME       SOURCE_KEY  DC_POWER  AC_POWER  ...  TOTAL_YIELD  AMBIENT_TEMPERATURE  MODULE_TEMPERATURE  IRRADIATION
0 2020-05-15  1BY6WEcLGh8j5v7       0.0       0.0  ...    6259559.0            25.184316           22.857507          0.0
1 2020-05-15  1IF53ai7Xc0U56Y       0.0       0.0  ...    6183645.0            25.184316           22.857507          0.0
2 2020-05-15  3PZuoBAID5Wc2HD       0.0       0.0  ...    6987759.0            25.184316           22.857507          0.0
3 2020-05-15  7JYdWkrLSPkdwr4       0.0       0.0  ...    7602960.0            25.184316           22.857507          0.0
4 2020-05-15  McdE0feGgRqW7Ca       0.0       0.0  ...    7158964.0            25.184316           22.857507          0.0

[5 rows x 9 columns]
```

Fig.6